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(54) Title: ELECTRIC MACHINE STATOR WITH INNER COOLING CHANNELS

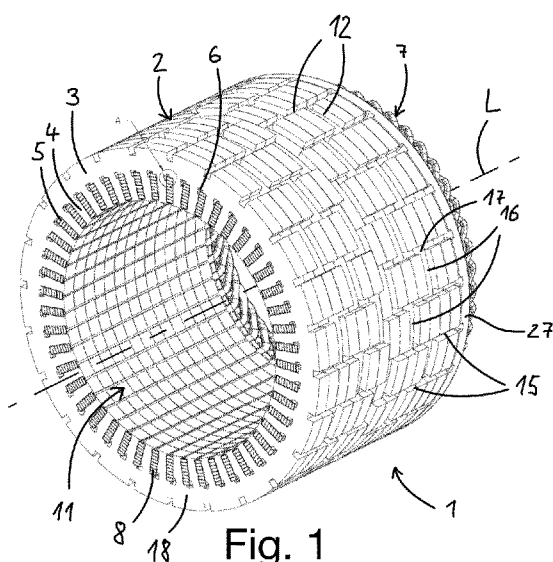


Fig. 1

(57) Abstract: An electric machine stator has an axis and comprises a stator core having a back iron portion, a plurality of teeth and a plurality of slots, the teeth extending radially inwards from the back iron portion with regard to the axis, alternating with the slots. A plurality of inner cooling channels extend through the stator core in parallel to the axis.

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Electric machine stator with inner cooling channels

Description

The application refers to an electric machine stator comprising a stator core having a back iron portion, a plurality of teeth and a plurality of slots.

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It is known that in the operation of electric machines such as generators or electric motors used to power automotive vehicles heat losses or copper losses occur. In order to avoid thermal overheating which would lead to lowered efficiency or even to damage or a reduction of the lifetime, particularly the insulation lifetime of the respective electric machine, cooling means are provided.

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For example, US 6,954,010 B1 discloses a device such as a motor, transformer or inductor that utilizes a stack of laminations, where a plurality of at least partially coincident apertures pass through the stack of laminations and define a plurality of coolant passageways. Manifold members located at opposite ends of the lamination stack are used to couple the coolant passageways to a suitable coolant pump and heat sink. A variety of aperture designs are disclosed, including both same-sized apertures that form straight passageways, and apertures that vary in size, shape and/or position to form non-axial passageways.

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US 10,128,701 B1 discloses an electric motor cooling system in which a plurality of axial coolant channels is integrated into the stator teeth. The axis of each of the axial coolant channels is parallel with the axis of the stator. A coolant manifold assembly that is integrated into the stator fluidly couples the coolant channels within the stator to the source of coolant.

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US 2021/351642 A1 discloses an electric motor cooling system that utilizes stator-integrated axial coolant channels and a coolant manifold centrally located within the

stator to efficiently remove heat. The coolant manifold includes a middle member that allows coolant to enter the axial coolant channels via transition laminations, where the transition laminations include coolant distribution channels that direct the flow of coolant entering into the manifold into the stator-integrated axial coolant channels.

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US 2022/0014062 A1 discloses an electric machine including a rotor and at least one cooling channel. The rotor is rotatable about an axis defining an axial direction and includes a stator having electrically conductive stator windings. A coolant flows through at least one cooling channel to cool the stator windings. The stator includes teeth extending along the axial direction. The cooling channel and the stator windings are arranged in an intermediate space formed between two adjacent stator teeth. A plastic for transmitting heat from the stator windings to the cooling channel is arranged in the intermediate space.

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US 2022/0200382 A1 discloses in a stator stack an in-slot cooling system that includes: two or more conductors of a hairpin winding provided within each slot in a first group of the slots, each conductor running axially across the stator stack and being at a different predetermined radial distance from an air gap; and adjacent each slot of the first group of slots at the end that is in the vicinity of the air gap, a bridge portion of the laminating material that closes the slot and forms in the stator stack substantially adjacent the conductor having the shortest predetermined distance from the air gap an axial cooling channel that allows a coolant to flow therein without leakage therefrom.

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An objective can be to provide an electric machine stator with improved flow-path of the cooling fluid.

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According to a first aspect, the objective is achieved by the electric machine stator according to claim 1. The electric machine stator has an axis and comprises a stator core having a back iron portion, a plurality of teeth and a plurality of slots, the teeth extending radially inwards from the back iron portion with regard to the axis, alternating with the slots. A plurality of inner cooling channels extend through the stator core in parallel to the axis. At least one of the inner cooling channels is provided as an integrated cooling channel, the integrated cooling channel extending at least partly in one of the teeth and being interconnected to one of the slots.

Interconnecting the integrated cooling channel to the slot while extending at least partly in the adjacent tooth is an advantageous enhancement, compared to channels integrated either in the teeth or in the slots, only. Channels in the teeth have a strong limitation with regard to the circumferential width due to the small circumferential extension of the teeth. Channels in the slots have less contact surface to the stator core, which retains a significant amount of the heat to be dissipated. The integrated cooling channels interconnected to the slots allow various locations and cross-sections, having for example a lower aspect ratio of the circumferential and radial extension than channels extending in the teeth only. The integrated cooling channel is interconnected to the slot and at least partly extending in the tooth when the voids forming the integrated cooling channel and the slots are connected or partly overlap, but do not fully overlap, in the sense that a material of the stator core does not separate the integrated cooling channel from the slot, but the integrated cooling channel extends beyond the slot, in particular in circumferential direction. For example, one slot and one or more integrated cooling channels are formed as one shared void. Further, one or two integrated cooling channels may be provided per each pair of slots and teeth.

According to an embodiment, the integrated cooling channels are arranged in a transition region between the back iron portion and the slots, i.e. at radially outward ends of the slots. The integrated cooling channel may extend in part in one of the teeth and in part in the back iron portion. Additionally, certain inner cooling channels not being integrated cooling channels can be arranged at radially inward ends of the slots. According to a further embodiment, the integrated cooling channels can be arranged radially extending along one side or both sides of the slots.

According to a further embodiment, the integrated cooling channels and optionally the inner cooling channels have impermeable conduits fluidly separating a flow of cooling fluid inside the conduits from the slots. The impermeable conduits can be made of a plastic composite material. The plastic composite material of the impermeable conduits can extend into the slot, closing the slot towards an inner diameter of the stator core, for example, if the slots are open towards the inner diameter and the inner cooling channels are arranged at radially inward ends of the slots.

According to a further embodiment, the electric machine stator further comprises a winding having a plurality of conductors, the conductors extending through the slots in parallel to the axis, an insulation encasing the conductors, wherein the insulation is integrally formed with the impermeable conduits.

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According to a second aspect, the objective is achieved by the electric machine stator according to claim 7. The electric machine stator has an axis and comprises a stator core having a back iron portion, a plurality of teeth and a plurality of slots, the teeth extending radially inwards from the back iron portion with regard to the axis, alternating with the slots. A plurality of inner cooling channels extend through the stator core in parallel to the axis and a plurality of outer cooling channels extending radially outward of the inner cooling channels through the stator core, the outer cooling channels being fluidly connected to the inner cooling channels via radially extending connecting channels, wherein the connecting channels are arranged at a distal axial end of the stator core. The inner cooling channels may partly or completely be provided as the integrated cooling channels described in respect of the first aspect.

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Advantageously, the plurality of inner cooling channels, outer cooling channels and connecting channels form a flow path for a cooling fluid, the cooling fluid flowing through the outer cooling channels towards the distal axial end of the stator core, radially inwards along the distal axial end of the stator core through the connecting channels, and into inner cooling channels extending through the stator core in parallel to the axis. A longer flow path through the stator core is provided by the proposed flow path.

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According to an embodiment, the outer cooling channels extend at a radially outward end of the back iron portion. The connecting channels can be arranged at both distal axial ends of the stator core, advantageously allowing opposite flow directions in adjacent inner cooling channels. The outer cooling channels can be formed as open grooves in a radially outer surface of the stator core. The outer cooling channels may be closed by a housing of the stator.

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According to a further embodiment, the outer cooling channels have axially extending sections and circumferentially extending sections, forming together a branched pattern

of the outer cooling channels.

According to a further embodiment, the stator core is formed of a plurality of laminations stacked in axial direction, wherein a first type of laminations has circumferentially distributed notches in a radially outer surface, the notches of the first type of laminations forming the axially extending sections of the outer cooling channels when stacked, and wherein a second type of laminations has circumferentially distributed openings in the radially outer surface, the openings spanning a circumferential distance overlapping two adjacent notches of the first type of laminations, the openings of the second type of laminations forming the circumferentially extending sections of the outer cooling channels when stacked. In the stack of laminations, for example, two laminations of the first type are followed by one second type lamination, each opening connecting two adjacent notches.

According to further embodiments, at least a part of the inner cooling channels may be fully integrated into the slots and/or the inner cooling channels may be arranged at radially inward ends of the slots and/or two or more inner cooling channels may be provided per slot.

Exemplary embodiments are illustrated in context of the accompanying drawings. In the Figures

Figure 1 shows an embodiment of an electric machine stator in a perspective view;

Figure 2 shows a detail of Figure 1;

Figure 3 shows the detail of Figure 2 in a different illustration;

Figure 4 shows an end cap of the stator core of Figure 1 in a perspective view;

Figure 5 shows the end cap of Figure 4 in a further perspective view;

Figures 6a and 6b show an electric machine with the electric machine stator;

Figure 7 shows a first type of lamination forming the electric machine stator of Figure 1;

Figure 8 shows a detail of the lamination of Figure 7;

Figure 9 shows a second type of lamination forming the electric machine stator of Figure 1;

Figure 10 shows a detail of the lamination of Figure 9;

Figure 11 shows a further embodiment of the electric machine stator;

Figure 12 shows yet another embodiment of the electric machine stator;

Figure 13 shows yet another embodiment of the electric machine stator;

Figure 14 shows yet another embodiment of the electric machine stator;

Figure 15 shows yet another embodiment of the electric machine stator.

Figure 1 shows an embodiment of an electric machine stator 1 in a perspective and partly cut view. Figure 2 and Figure 3 show a detail A of Figure 1 in two different illustrations. The Figures 1 through 3 are described together. The electric machine stator 1 has an axis L, which may as well be referred to as a longitudinal axis L. It defines an axial direction. In an assembled electric machine comprising the stator 1, a not depicted rotor is arranged in an inner diameter 11 of the stator 1, rotatable about the axis L. A stator core 2 has a back iron portion 3, a plurality of teeth 4 and a plurality of slots 5, the teeth 4 extending radially inwards from the back iron portion 3 with regard to the axis L, alternating with the slots 5. The stator core 2 can be formed of a plurality of insulated ferromagnetic iron laminations 18 stacked in the axial direction. The person skilled in the art is aware that the stator core 2 may alternatively be formed as a spirally wound insulated ferromagnetic iron strip. The electric machine stator 1 further com-

prises a winding 7, for example a distributed winding 7, a winding head of which extends out of an end cap 27 at a far end of the stator 1 in the perspective view. The end cap 27 and the winding 7 is not depicted at the near end of the stator 1, where a sectional view through a plurality of conductors 8 of the winding 7 is shown. The conductors 8 extending through the slots 5 in parallel to the axis L are not depicted as individual wires, which may be inserted groups of wire coils or plug-in windings, like hairpin windings. The detail A in Figure 2 shows an insulation 10 encasing the conductors 8.

A plurality of inner cooling channels are provided as integrated cooling channels 6' according to a first aspect. The integrated cooling channels 6' extend through the stator core 2 in parallel to the axis L. Each of the integrated cooling channels 6' is interconnected to one of the slots 5, i.e. the integrated cooling channels 6' and the slots 5 are formed inside a common void, as illustrated in Figure 3 showing the detail A without the insulation 10 of the conductors 8. The material of the stator core 2 does not separate the integrated cooling channels 6' from the slots 5. In the depicted embodiment, the integrated cooling channels 6' are arranged in a transition region between the back iron portion 3, the teeth 4 and the slots 5. The integrated cooling channels 6' extend partly in the teeth 4 and partly in the back iron portion 3. The stator core 2 further has outer cooling channels 12 at a radially outward end of the back iron portion 3. The outer cooling channels 12 can be formed as open grooves in a radially outer surface of the stator core 2. The outer cooling channels 12 may have axially extending sections 15 and circumferentially extending sections 16, the circumferentially extending sections 16 connecting two parallel axially extending sections 15, thus forming a channel pattern with multiple flow paths along the radially outward end of the back iron portion 3. In the depicted embodiment, one internal cooling channel 6' is provided per each pair of slots 5 and teeth 6.

According to a second aspect, the electric machine stator with the plurality of inner cooling channels 6 extending through the stator core 2 in parallel to the axis L and the plurality of outer cooling channels 12 extending radially outward of the inner cooling channels 6 through the stator core 2, the outer cooling channels 12 are fluidly connected to the inner cooling channels 6 via radially extending connecting channels 14. The connecting channels 14 are arranged at a distal axial end of the stator core 2. The

inner cooling channels 6 may partly or completely be provided as the integrated cooling channels 6' according to the first aspect. The end cap 27 is attached to both axially distal ends of the stator core 2, though only one of them is depicted in Figure 1. The Figures 4 and 5 show the end cap 27 in different perspective views, from the side facing the stator core 2 in Figure 5 and from the opposite side in Figure 4. The end caps 27 have the same slots 5 as the rotor core 2, but the integrated cooling channels 6' are only provided at every second slot 5, where the respective integrated cooling channels 6' are open towards the axial end of the stator core 2. The respective other half of the integrated cooling channels 6' is fluidly connected to the outer cooling channels 12 via the radially extending connecting channels 14 formed in the end caps 27. The same end cap 27 is arranged at both axial distal ends of the stator core 2, with the connecting channels 14 being rotated one slot pitch. Thus, the integrated cooling channels 6' are alternatingly open to the axial end of the stator core 2 and connected via the connecting channels 14 to the outer cooling channels 12.

Figures 6a and 6b show an electric machine in two different sectional views, which are described together. A housing 24 has the electric machine stator 1 of Figure 1 mounted therein. A rotor 29 is mounted on a shaft rotating about the axis L inside the electric machine stator 1. The open grooves forming the outer cooling channels 12 are closed by an inner surface 17 of the cylindrical part of the housing 24. A cooling fluid 30 like oil can be conveyed by a pump or by gravitational force to an inlet 25, which leads to the outer cooling channels 12. The two sections of Figures 6a and 6b are rotated relative to each other to show sectional views of two adjacent inner cooling channels 6, which may be provided as integrated cooling channels 6'. The cooling fluid 30 flows through the channel pattern with multiple flow paths formed by the outer cooling channels 12 towards both lateral ends of the stator core 2 and via the connecting channels 14 into the inner cooling channels 6, which may be provided as integrated cooling channels 6', and which are passed through by the cooling fluid 30 alternatingly in the one and the other axial direction, as illustrated by arrows B pointing in Figure 6a in one direction and in Figure 6b in the opposite direction. At the respective open ends, the cooling fluid 30 emerges from the inner or internal cooling channels 6, 6' and falls onto the winding heads of the winding 7, from where it reaches a bottom of the housing 24, where it is withdrawn from the housing 24 via at least one outlet 26, to be further conveyed, for example, to a not depicted heat exchanger. The plurality of inner cooling

channels 6 or integrated cooling channels 6', outer cooling channels 12 and connecting channels 14 form a flow path for the cooling fluid 30 coming from the inlet 25, the cooling fluid 30 flowing through the outer cooling channels 12 towards both distal axial ends of the stator core 2, radially inwards along the distal axial ends of the stator core 2 through the connecting channels 14, and into inner cooling channels 6 or integrated cooling channels 6' extending through the stator core 2 in parallel to the axis L.

The embodiment of the stator core 2 as shown in Figure 1 is formed of the stacked laminations 18, which are further described with regard to Figures 7 through 10, wherein Figure 7 shows a first type of laminations 20 forming the stator core 2 and Figure 9 shows a second type of laminations 22. The first type of laminations 20 has circumferentially distributed notches 21 in a radially outer surface, the notches 21 of the first type of laminations 20 forming the axially extending sections 15 of the outer cooling channels 12 when stacked. The second type of laminations 22 has circumferentially distributed openings 23 in the radially outer surface, the openings spanning a circumferential distance overlapping two adjacent notches 21 of the first type of laminations 20, the openings 23 of the second type of laminations 22 forming the circumferentially extending sections 16 of the outer cooling channels 12 when stacked. A circumferential distance B of the notches 21 may be for example 15°, whereas a circumferential distance D of the openings 23 may be 30°. Figure 8 shows a detail A of the first type lamination 20 of Figure 7, wherein the notches 21 may have circumferential width C of 6°. Figure 10 shows a detail A of the lamination of Figure 9, wherein the openings 23 may have circumferential width E of 21°, spanning two adjacent notches 21, as E equals B plus C.

Figures 11 through 14 show each a further embodiment of the electric machine stator 1 in a cross-sectional view with the axis L being orthogonal to the plane of projection. Only three adjacent slots 5 are depicted in a cut-out. The person skilled in the art is aware that the stator core 2 may have as many slots 5 as the embodiment of Figure 1, for example. One of the notches 21 forming the axially extending sections 15 of the outer cooling channels 12 is depicted, closed at the radially outer end by the housing 24.

The embodiment of Figure 11 has two integrated cooling channels 6' per slot 5 arranged in the transition region between the back iron portion 3 and the slots 5. Further, the integrated cooling channels 6' each have impermeable conduits 9 fluidly separating the flow of cooling fluid inside the conduits 9 from the slots 5. The impermeable conduits 9 can be made of a plastic composite material. The conductors 8 inside the slots 5 can have an insulation 10, for example an insulation paper. The slots 5 of the embodiment have an aperture 28 towards the inner diameter 11 of the stator core 2.

The embodiment of Figure 12 has one inner cooling channel 6 per slot 5, wherein the inner cooling channels 6 are fully received in the slots 5 and thus are not integrated cooling channels according to the first aspect. The inner cooling channels 6 are arranged at radially inward ends of the slots 5. The inner cooling channels 6 have the impermeable conduits 9 and the conductors 8 inside the slots 5 have an insulation 10. The plastic composite material of the impermeable conduits 9 extends into the aperture 28, closing the slot 5 towards the inner diameter 11 of the stator core 2.

The embodiment of Figure 13 has one inner cooling channel 6 per slot 5 fully received therein at radially inward ends of the slots 5. The inner cooling channels 6, which are not integrated cooling channels according to the first aspect, have the impermeable conduits 9 and the conductors 8 inside the slots 5 have an insulation 10, wherein the insulations 10 are integrally formed with the impermeable conduits 9. The plastic composite material of the impermeable conduits 9 extends into the aperture 28, closing the slot 5 towards the inner diameter 11 of the stator core 2.

The embodiment of Figure 14 has one inner cooling channel 6 per slot 5 fully received therein at radially inward ends of the slots 5. The inner cooling channels 6, which thus are not integrated cooling channels according to the first aspect, have the impermeable conduits 9 and the conductors 8 inside the slots 5 have an insulation 10, wherein the insulations 10 are integrally formed with the impermeable conduits 9. The slots 5 are closed towards the inner diameter 11 of the stator core 2.

The embodiment of Figure 15 has one inner cooling channel 6 per slot 5, which fully extends in the adjacent tooth 4 and which thus is not an integrated cooling channels according to the first aspect.

All features described in connection with the embodiments of Figures 1 and 11 through 15 can be transferred separately to any other embodiment.

Reference Numerals

1	Electric machine stator
2	Stator core
3	Back iron portion
4	Teeth
5	Slots
6	Inner cooling channels
6'	Integrated cooling channels
7	Winding
8	Conductors
9	Impermeable conduits
10	Insulation
11	Inner diameter
12	Outer cooling channels
14	Connecting channels
15	Axially extending sections
16	Circumferentially extending sections
17	Inner surface
18	Laminations
20	First type of laminations
21	Notches
22	Second type of laminations
23	Openings
24	Housing
25	Inlet
26	Outlet
27	End caps
28	Aperture

29	Rotor
30	Cooling fluid
B	Arrows
L	Axis

Electric machine stator with inner cooling channels

Claims

1. An electric machine stator (1) having an axis (L) and comprising:

a stator core (2) having a back iron portion (3), a plurality of teeth (4) and a plurality of slots (5), the teeth extending radially inwards from the back iron portion with regard to the axis, alternating with the slots,

a plurality of inner cooling channels (6) extending through the stator core (2) in parallel to the axis (L),

characterised in

that at least one of the inner cooling channels (6) is provided as an integrated cooling channel (6'), the integrated cooling channel extending at least partly in one of the teeth (4) and being interconnected to one of the slots (5).
2. The electric machine stator according to claim 1, characterised in that the at least one integrated cooling channel (6') is arranged in a transition region between the back iron portion (3) and the slots (5).
3. The electric machine stator according to any one of the proceeding claims, characterised in that the inner cooling channels (6) have impermeable conduits (9) fluidly separating a flow of cooling fluid inside the conduits from the slots (5).
4. The electric machine stator according to claim 3, characterised in that the impermeable conduits (9) are made of a plastic composite material.

5. The electric machine stator according to claim 4, characterised in that the plastic composite material of the impermeable conduits (9) extends into an aperture (28), the composite material in the aperture closing the slot (5) towards an inner diameter (11) of the stator core (2).
6. The electric machine stator according to any one of the proceeding claims 3 or 4, characterised in that the electric machine stator further comprises a winding (7) having a plurality of conductors (8), the conductors extending through the slots (5) in parallel to the axis (L), with an insulation (10) encasing the conductors, wherein the insulation is integrally formed with the impermeable conduits (9).
7. An electric machine stator (1), preferably according to any one of the proceeding claims, having an axis (L) and comprising:
 - a stator core (2) having a back iron portion (3), a plurality of teeth (4) and a plurality of slots (5), the teeth extending radially inwards from the back iron portion with regard to the axis (L), alternating with the slots,
 - a plurality of inner cooling channels (6) extending through the stator core (2) in parallel to the axis (L),
 - a plurality of outer cooling channels (12) extending radially outward of the inner cooling channels (6) through the stator core (2),
 - wherein the outer cooling channels (12) are fluidly connected to the inner cooling channels (6) via radially extending connecting channels (14),
 - characterised in
 - that the connecting channels (14) are arranged at a distal axial end of the stator core (2).

8. The electric machine stator according to claim 7, characterised in that the plurality of inner cooling channels (6), outer cooling channels (12) and connecting channels (14) form a flow path for a cooling fluid (30), the cooling fluid flowing through the outer cooling channels (12) towards the distal axial end of the stator core (2), radially inwards along the distal axial end of the stator core (2) through the connecting channels (14), and into inner cooling channels (6) extending through the stator core (2) in parallel to the axis (L).
9. The electric machine stator according to any one of the proceeding claims 7 or 8, characterised in that the outer cooling channels (12) extend through a radially outward range of the back iron portion (3).
10. The electric machine stator according to any one of the proceeding claims 7 to 9, characterised in that the connecting channels (14) are arranged at both distal axial ends of the stator core (2).
11. The electric machine stator according to any one of the proceeding claims 7 to 10, characterised in that the outer cooling channels (12) are formed as open grooves in a radially outer surface of the stator core (2).
12. The electric machine stator according to any one of the proceeding claims 7 to 11, characterised in that the outer cooling channels (12) have axially extending sections (15) and circumferentially extending sections (16), forming a branched pattern of the outer cooling channels.
13. The electric machine stator according to any one of the proceeding claims 7 to 12, characterised in that the stator core is formed of a plurality of laminations (18) stacked in axial direction, wherein a first type (20) of laminations has circumferentially distributed notches (21) in a radially outer surface, the notches of the first type of laminations forming the axially extending sections (15) of the outer cooling channels (12) when stacked, and wherein a second type (22) of laminations has circumferentially distributed openings (23) in the radially outer surface, the openings spanning a circumferential distance overlapping two adjacent notches (21) of the first type (20) of

laminations, the openings (23) of the second type of laminations forming the circumferentially extending sections (16) of the outer cooling channels (12) when stacked.

14. The electric machine stator according to any one of the proceeding claims 7 to 13, characterised in that the inner cooling channels (6) are fully integrated into the slots (5).
15. The electric machine stator according to any one of the proceeding claims 7 to 14, characterised in that the inner cooling channels (6) are arranged at radially inward ends of the slots (5).

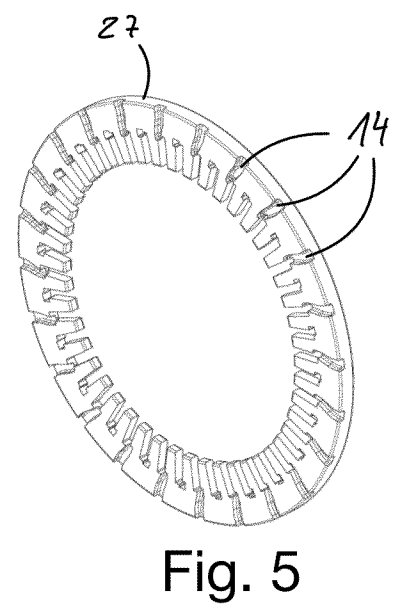
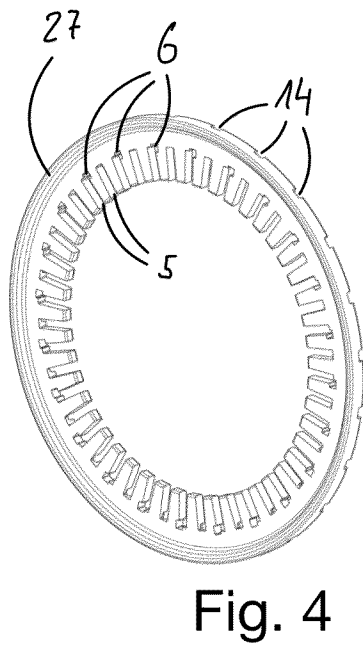
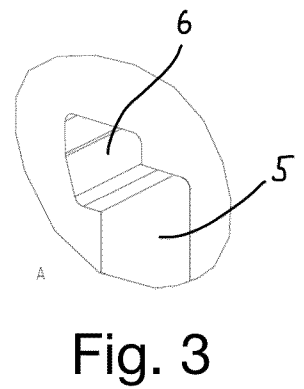
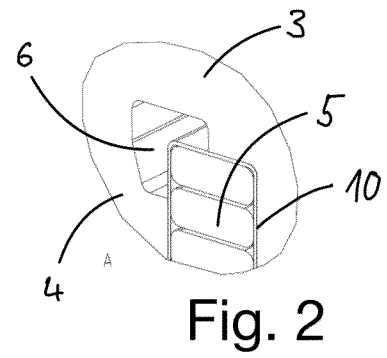
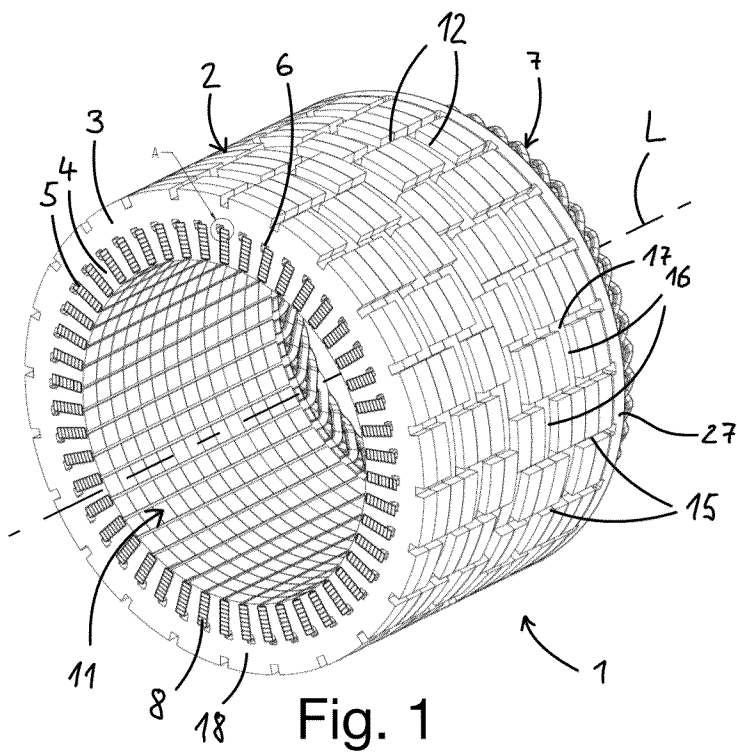


Fig. 6a

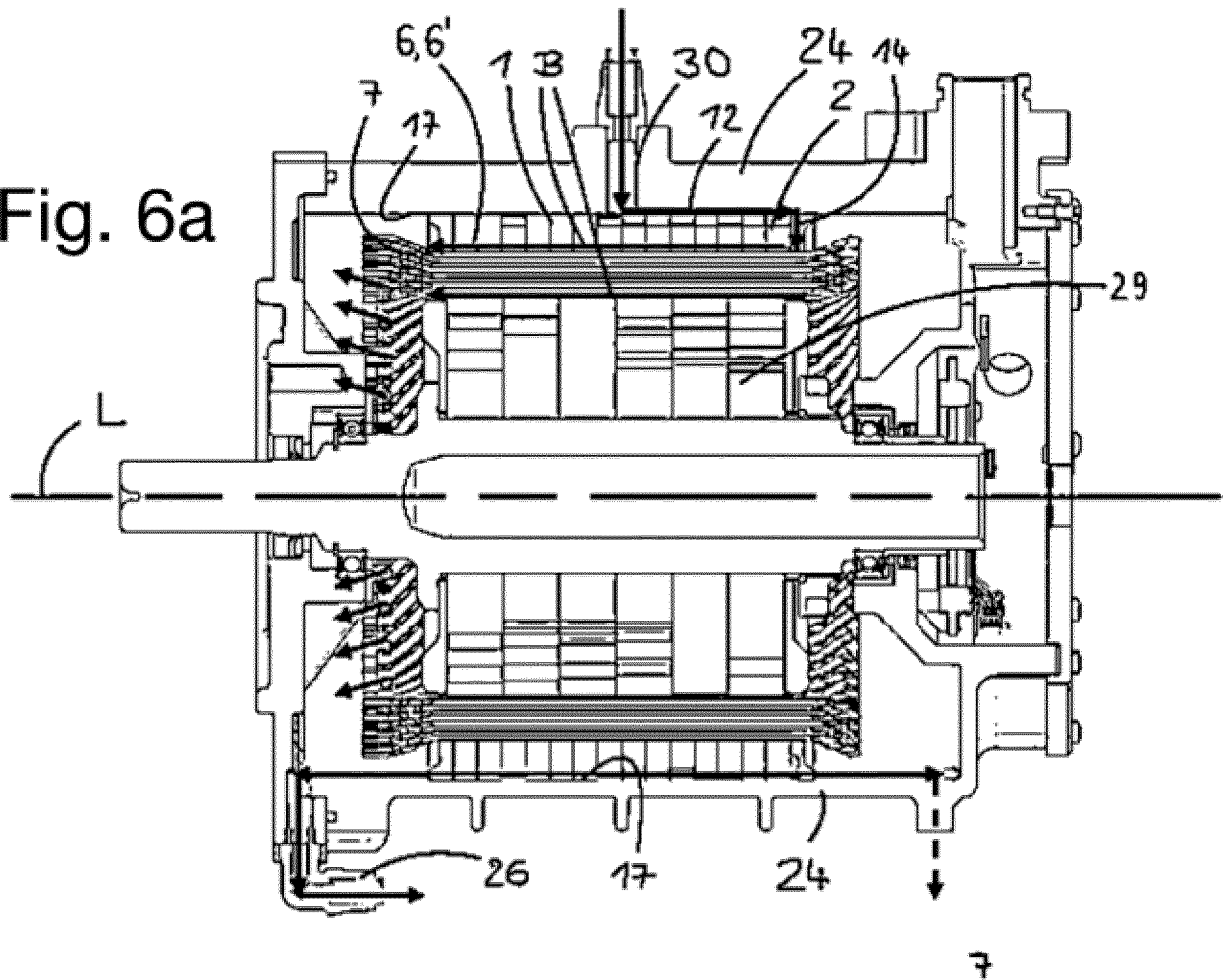
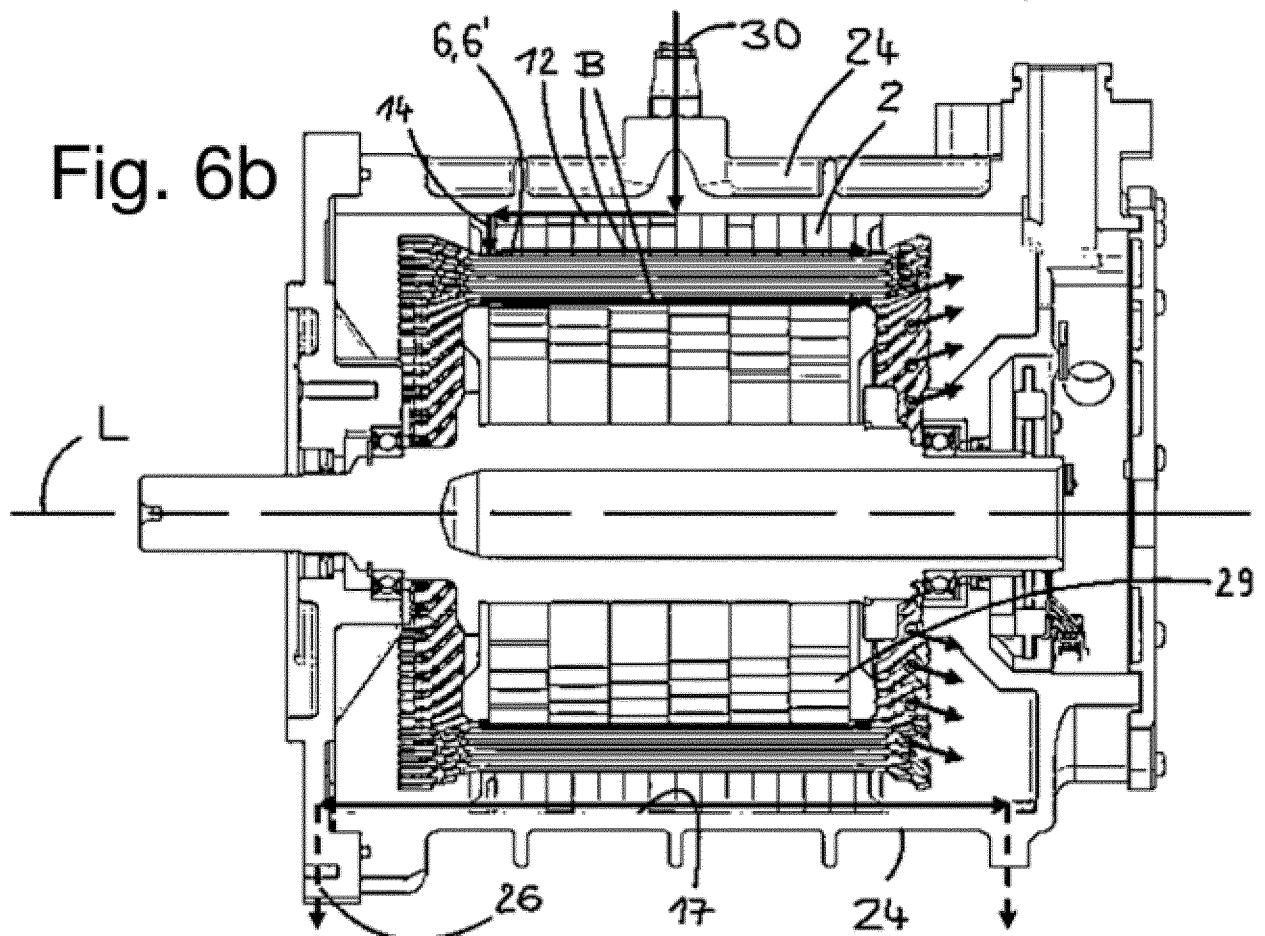


Fig. 6b



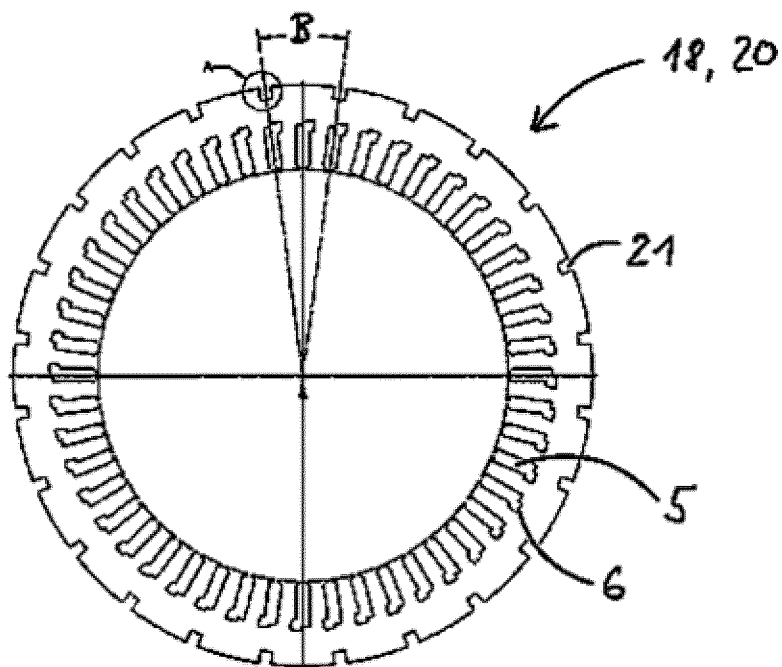


Fig. 7

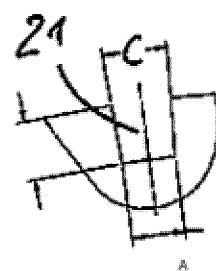


Fig. 8

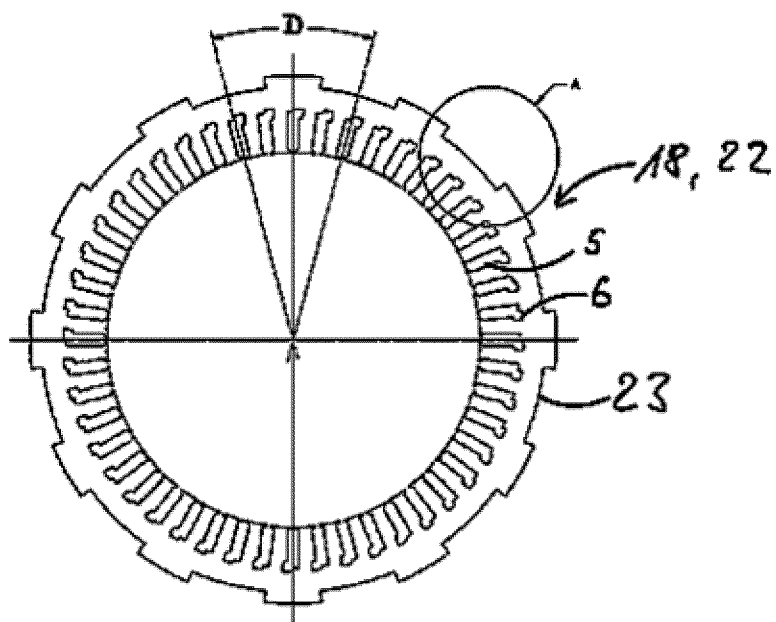


Fig. 9

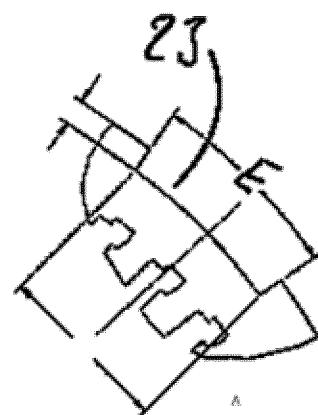
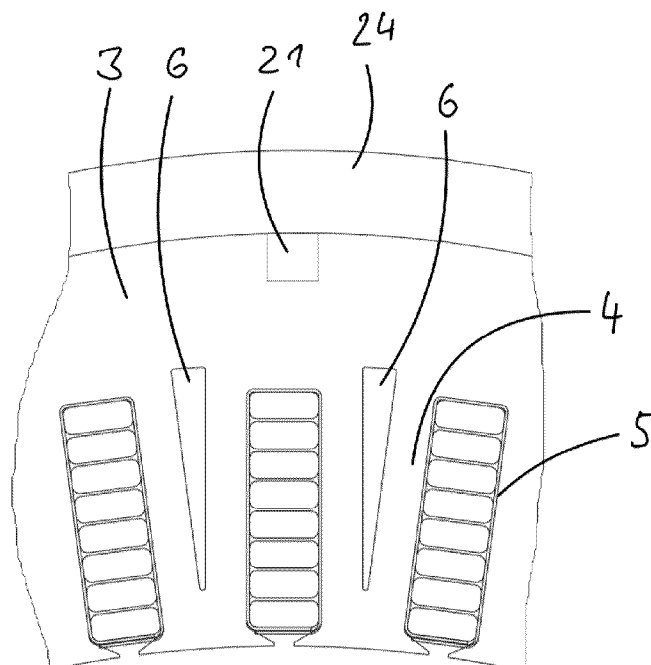
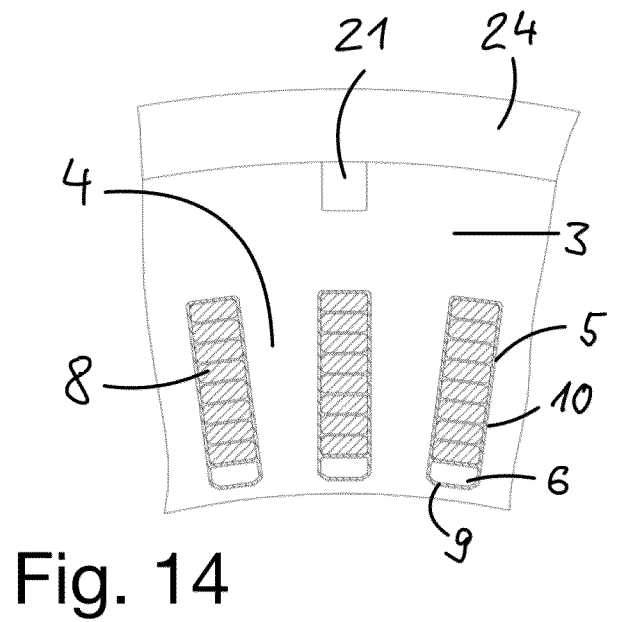
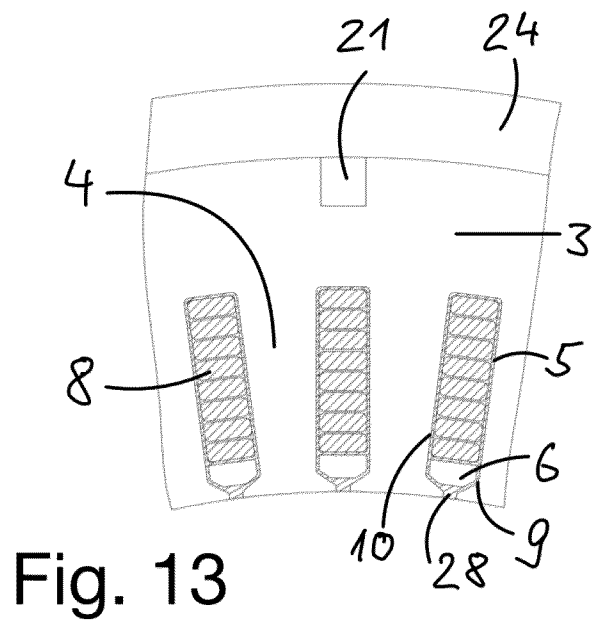
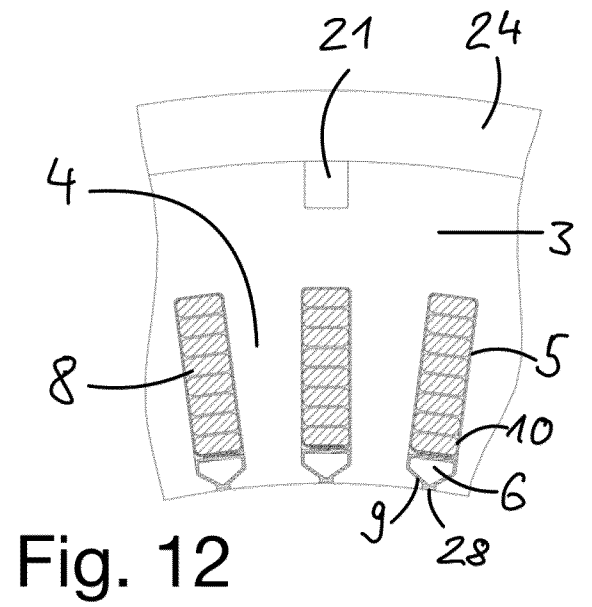
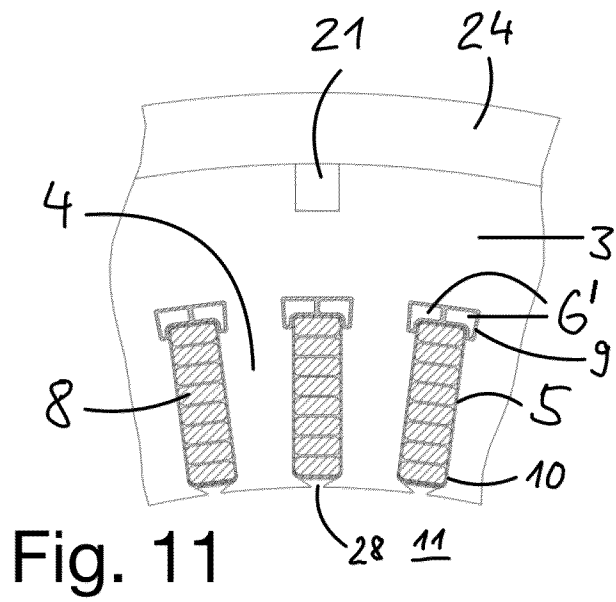


Fig. 10



INTERNATIONAL SEARCH REPORT

International application No

PCT/EP2022/074252

A. CLASSIFICATION OF SUBJECT MATTER

INV. H02K1/20 H02K3/24 H02K9/19
ADD.

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)

H02K

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)

EPO-Internal

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
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Y	page 11, line 29 - page 12, line 7 claim 1 figure 8C	3-6
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X	DE 10 2020 107165 A1 (SCHAEFFLER TECHNOLOGIES AG [DE]) 29 July 2021 (2021-07-29) paragraph [0015] figure 2	1
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Further documents are listed in the continuation of Box C.



See patent family annex.

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Date of the actual completion of the international search

6 April 2023

Date of mailing of the international search report

21/04/2023

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Maas, Erik

INTERNATIONAL SEARCH REPORT

International application No

PCT/EP2022/074252

C(Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT		
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INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No

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